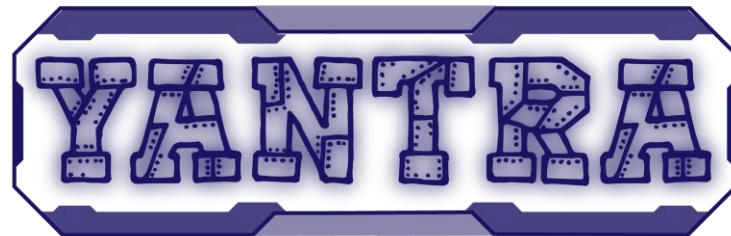
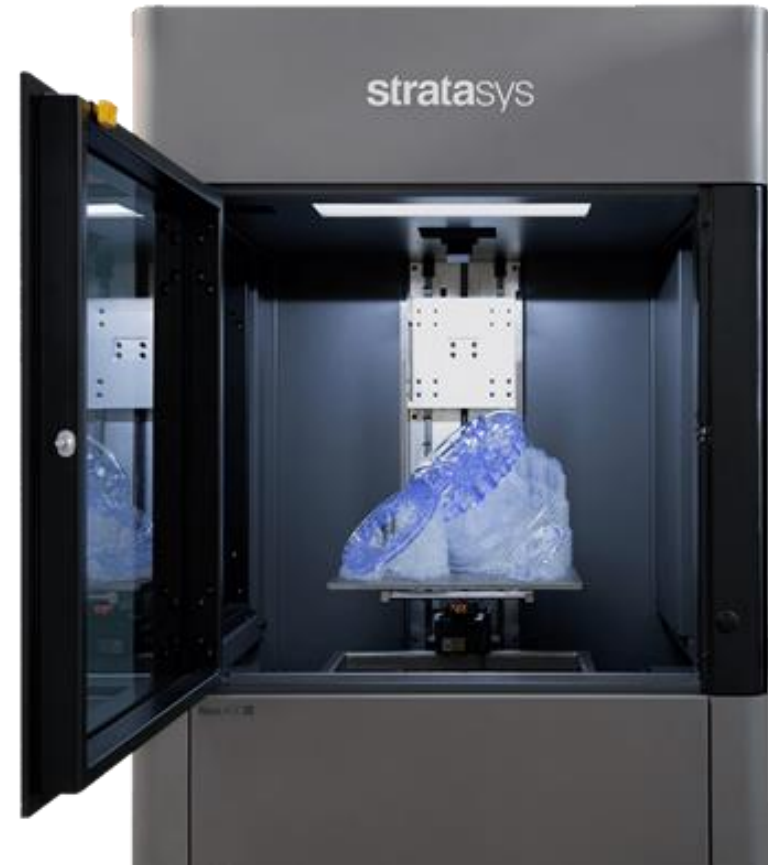


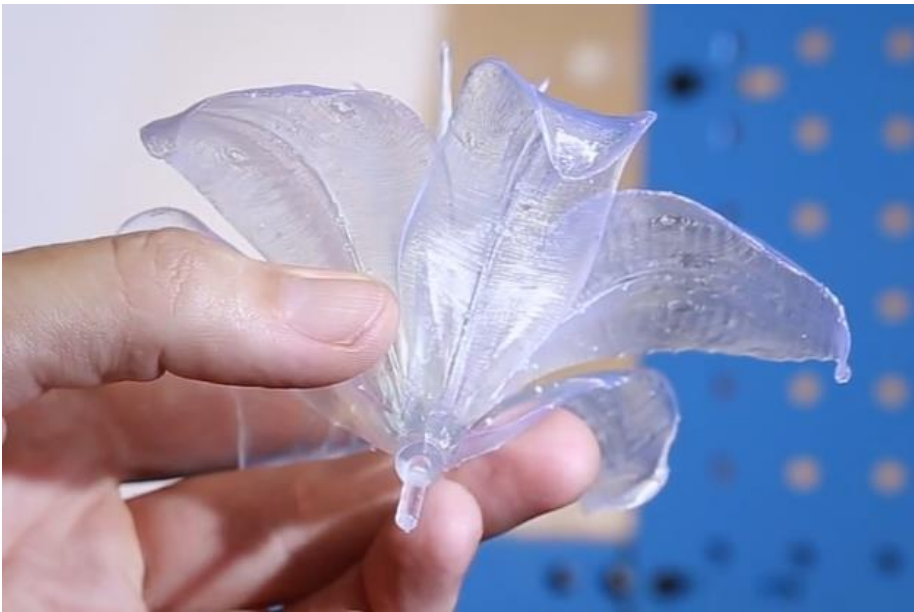
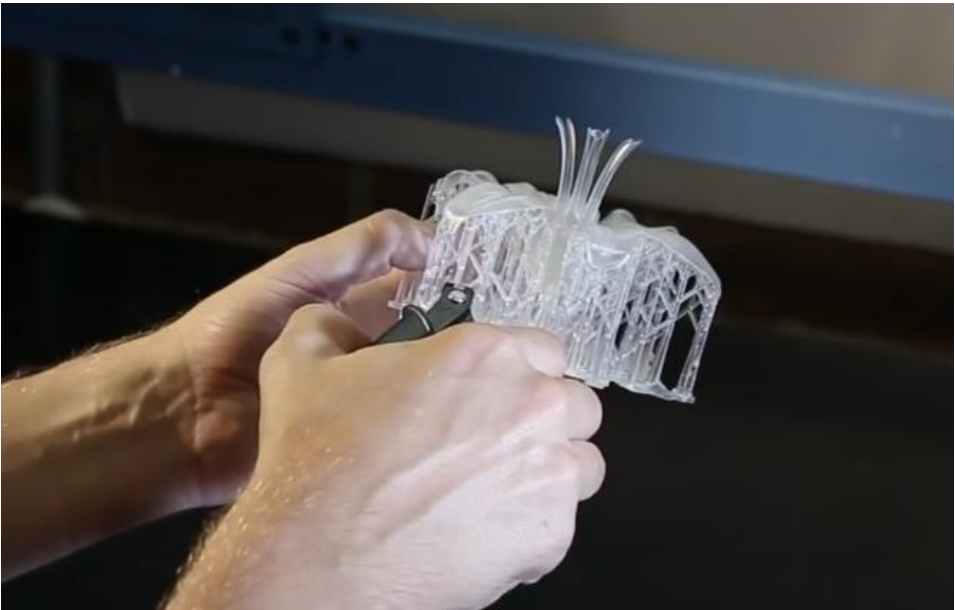
Stereolithography



Stereolithography

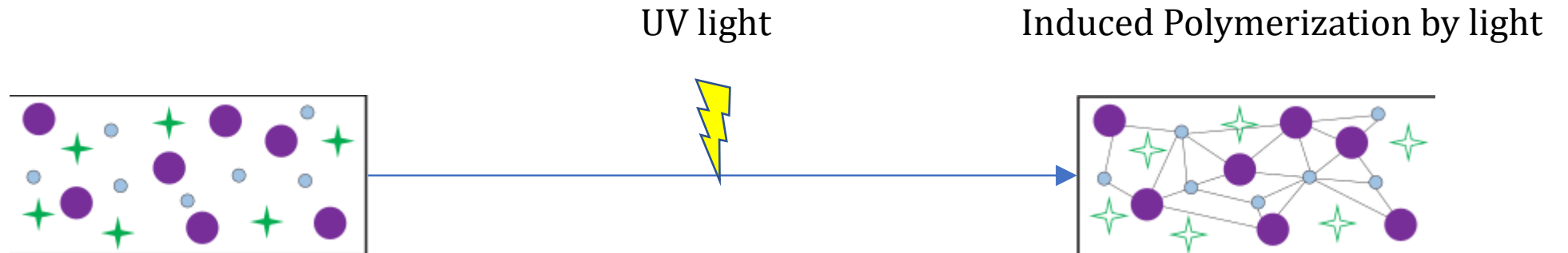


Printers



Principle

- The SLA process is based fundamentally on the following principles:
 1. Parts are built from a **photo-curable liquid resin** that cures when exposed to a **laser beam** (basically, undergoing the photo polymerization process) which scans across the surface of the resin.
 2. The building is done layer by layer, each layer being scanned by the **optical scanning system** and controlled by an elevation mechanism which lowers at the completion of each layer.



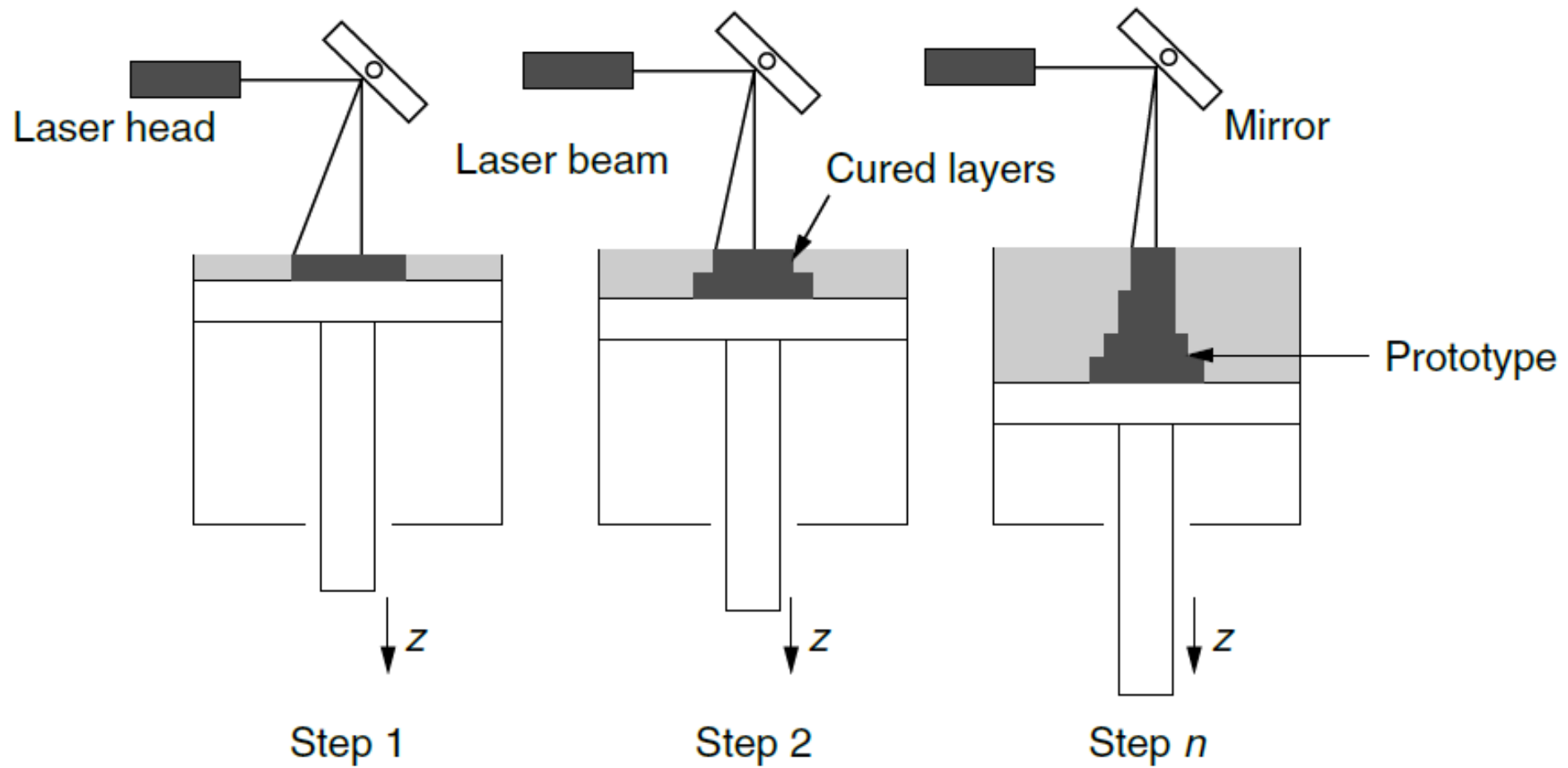
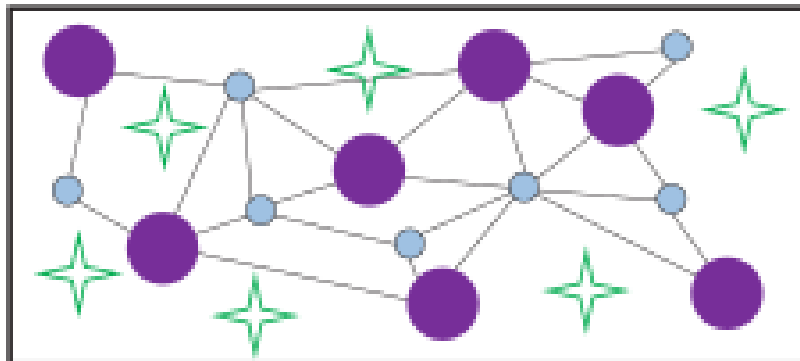


Figure 8.1 Principles of stereolithography.

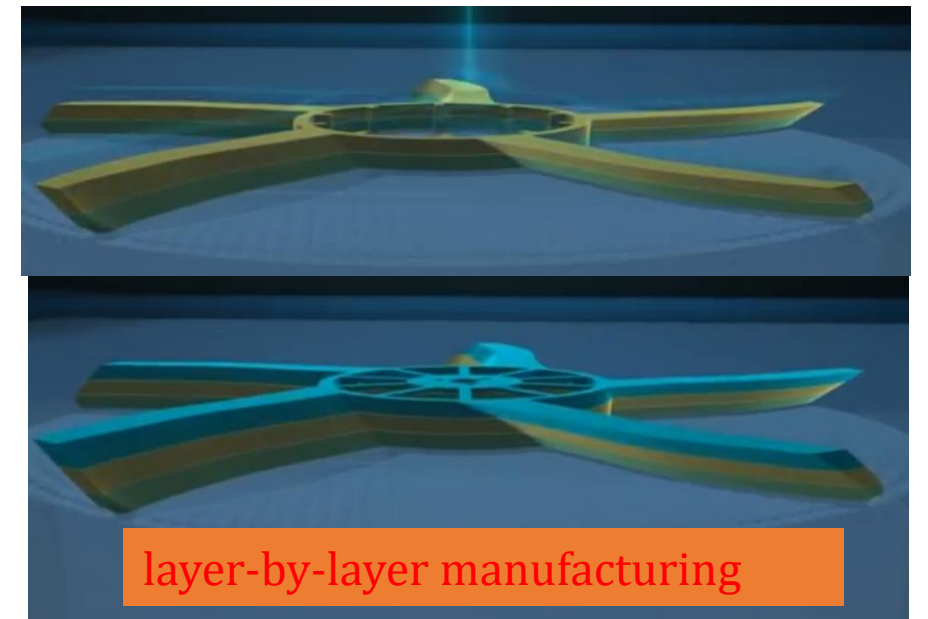
- Polymerization is the process of linking small molecules (known as **monomers**) into chain-like larger molecules (known as **polymers**).
- When the chain-like polymers are linked further to one another, a cross-linked polymer is said to be formed.



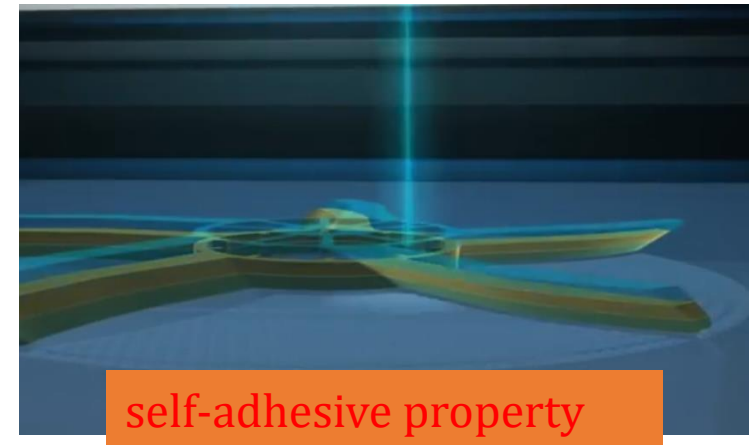
a **photopolymer** is a light-sensitive material that changes physical properties when exposed to light.



- Stereolithography (SLA) is the most common RP technique that employs **layer-by-layer manufacturing** based on photopolymerization.
- Solidification happens by **photopolymerization** that results from the impact of a **laser** light on the upper surface of the liquid.
- **Light polymerization** of the liquid is limited to a layer thickness of a **few tenths of a millimeter** below the surface.



- Illumination and hence polymerization of the surface liquid is restricted to a pattern that corresponds to the part's cross section.
- Once the **layer** is completely traced,
- it is **lowered by a small distance** into the liquid
- and a second layer is traced right on top of the first.
- The **self-adhesive property** of the material causes the layers to bond to one another
- and eventually form a complete, **three-dimensional object**



The following is the **sequence** of the SLA process:

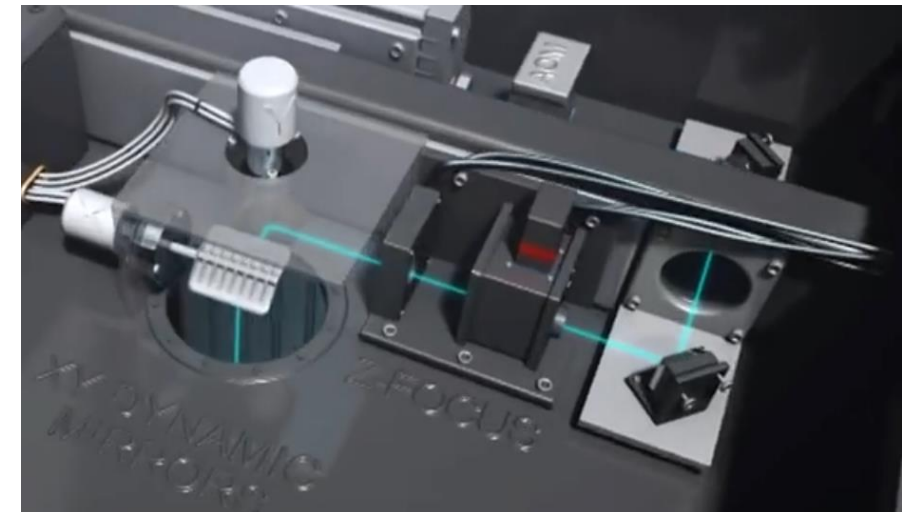
1. Slicing software **reads the CAD** model and **cuts it into thin slices** that depend on the resolution of the mechanical movements of the RP system as well as on the part geometry.



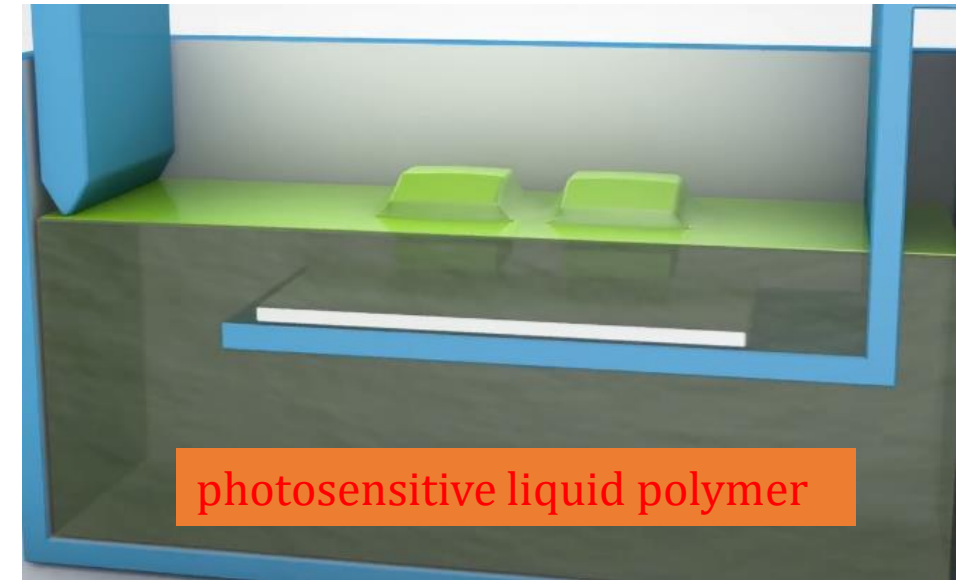
2. The computer receives the **output from the slicing software** that controls the corresponding **transitions and rotations** of the mechanical components of the RP system.



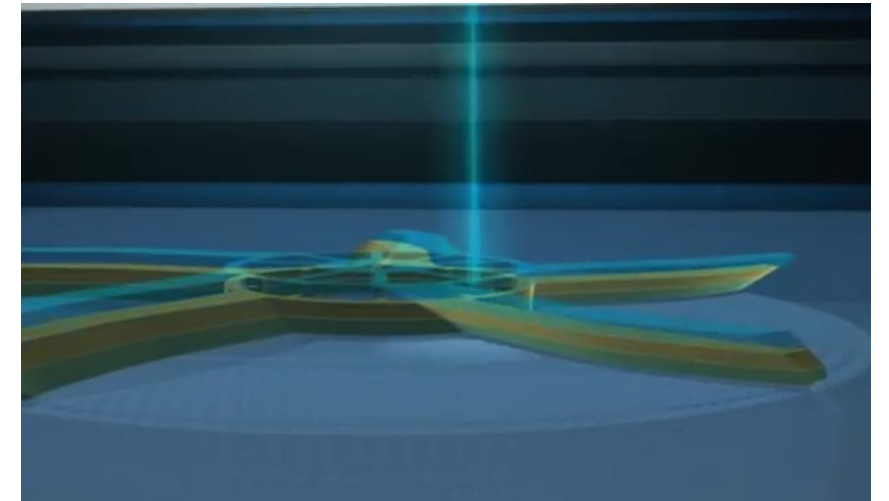
**mechanical
components of the RP
system.**



3. The process chamber filled with the **photosensitive liquid polymer** has an elevator-like platform that can be raised vertically or lowered.

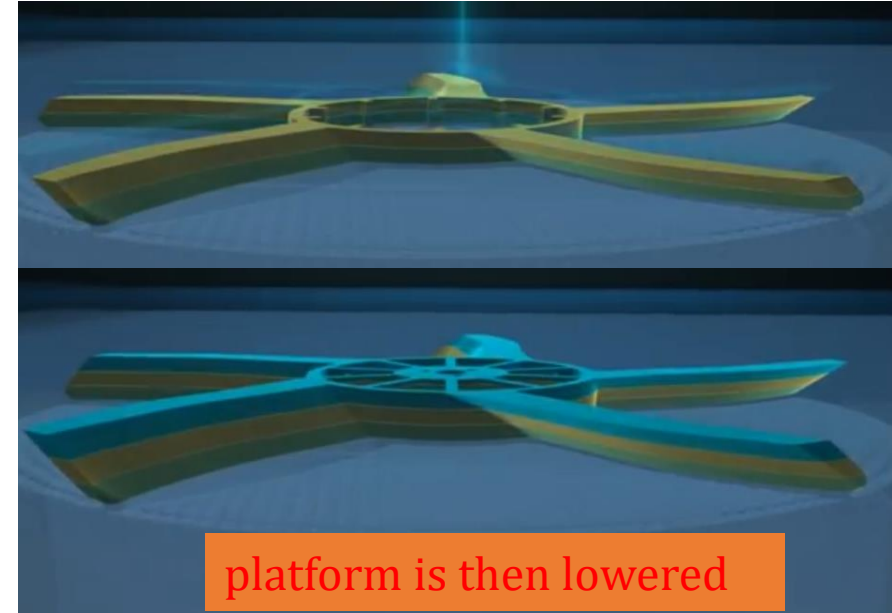


4. The laser beam **scans the selected surface area** of the photopolymer. It cures the scanned patterns and produces a solid section.



laser beam scans the selected surface area

5. The platform is then **lowered** so that another shallow layer of the liquid polymer covers the previously cured layer.



6. The sequence is **repeated** to produce the complete part geometry.





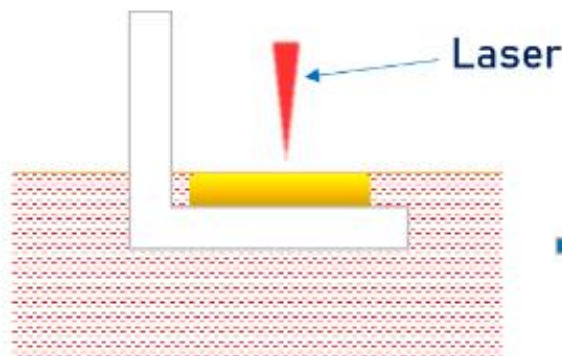
Data by 3D CAD



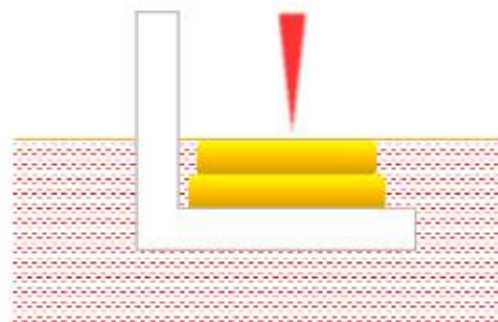
CAD data to STL
file format



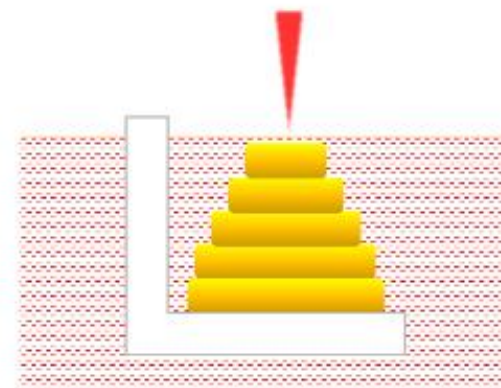
Slicing to N
layers



Hardening 1st
layer



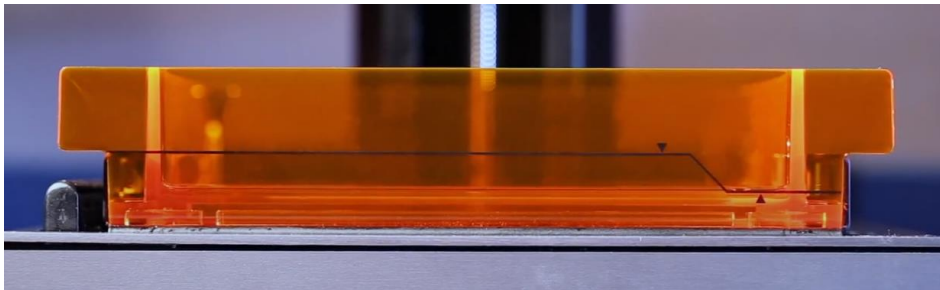
Hardening 2nd
layer



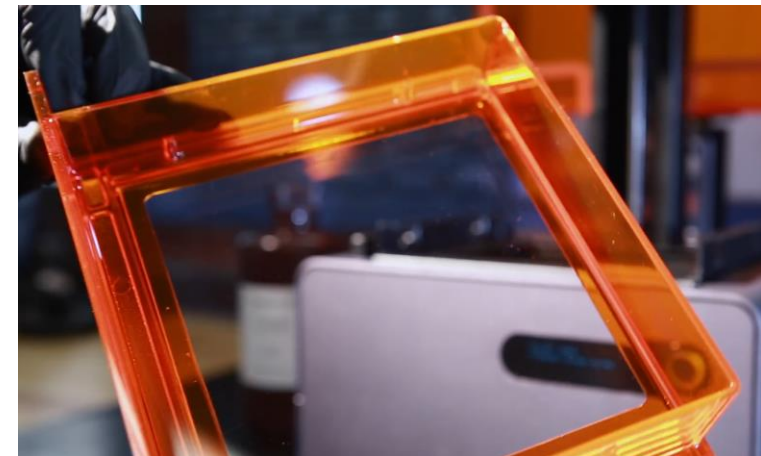
Hardening final
layer

Hardware

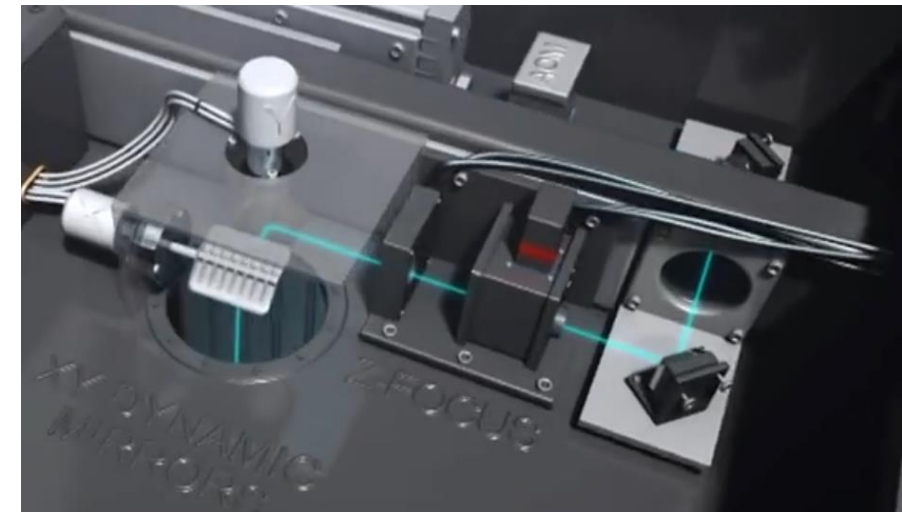
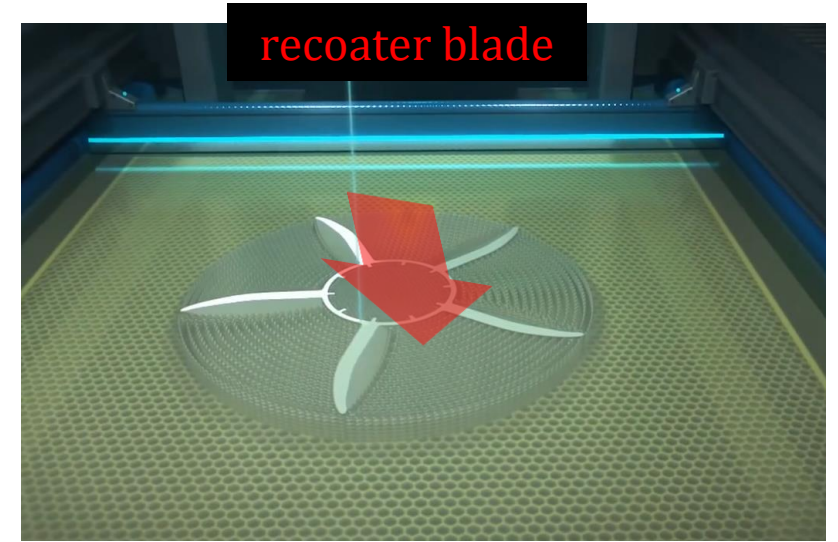
- The build chamber of the SLA contains a removable vat that holds the build resin, a detachable, perforated build platen on a -z axis elevator frame,
- and an automated resin-level checking apparatus. The vat has a small amount of -z movement capability, which allows the computer to maintain the exact height per layer.



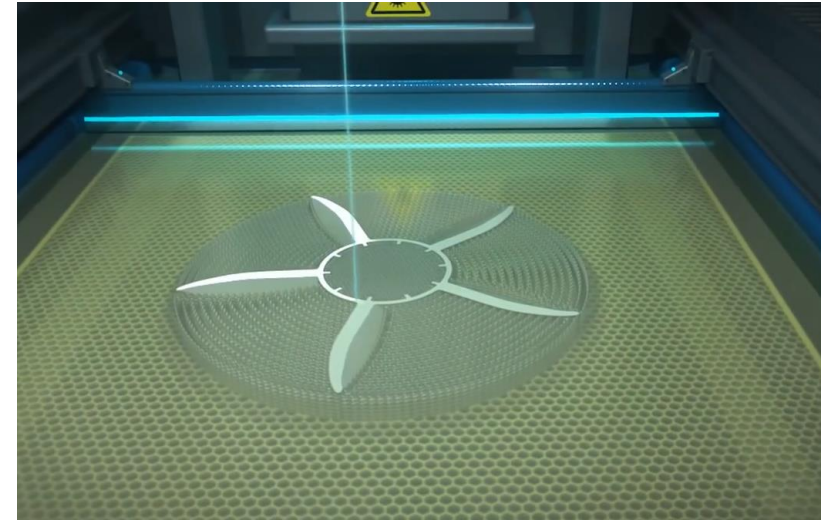
removable vat



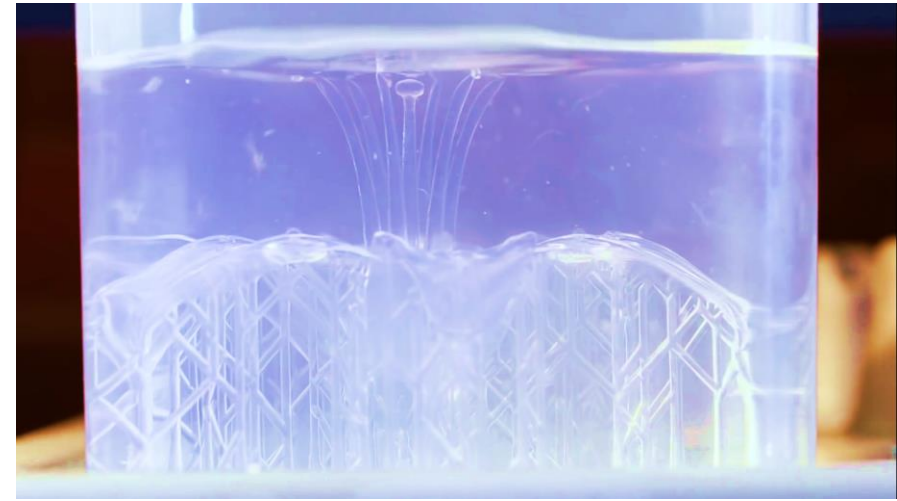
- In an enclosed area above and behind the build chamber, resides the laser and optics required to cure the resin.
- The laser unit is long and rectangular, about 4 feet long, and remains stationary.
- The laser beam is transferred to the part surface below by a series of optics, the final of which moves to scan the cross section of the part being built



- A recoater blade rides along a track at the top of the vat, and serves to "smooth" the liquid across the part surface to prevent a rounding of edges due to cohesion effects.
- Some systems now have a Zephyr recoater blade, which actually siphons up resin and delivers it evenly across the part surface



- Post processing units: an ultraviolet oven call the Post Curing Apparatus (PCA);
- and an alcohol bath large enough to hold entire build platens with parts attached.
- Parts are washed in the alcohol or a similar solvent immediately after being removed from the machine (while still attached to the build platen).
- This step removes any extra resin that clings to the surfaces of the part.
- After the final supports are removed, with some build styles the parts are required to be placed in the PCA to finish fully curing.



PROCESS PARAMETERS

1. Laser beam spot diameter:

The laser beam diameter is fixed at **200 μm** on the build surface for most of the SLA systems.

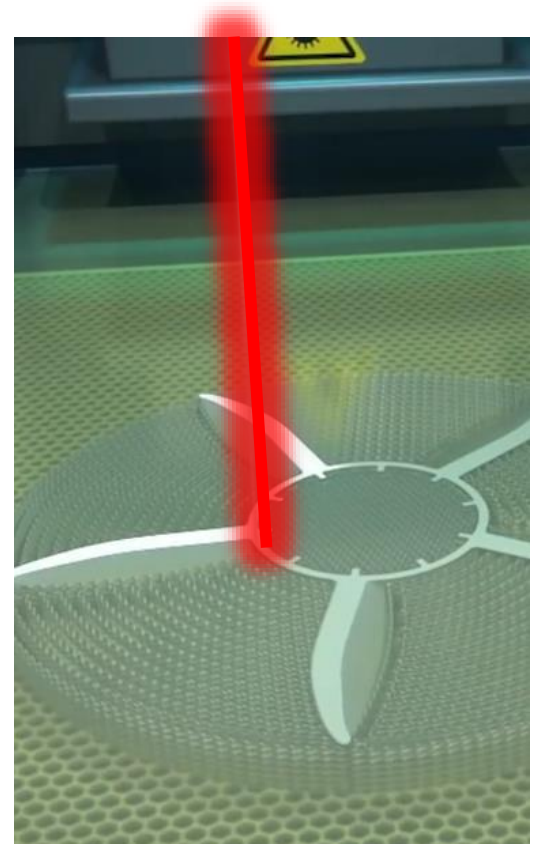
In theory this is the absolute limiting size of a feature that can be built.

However, because of the scanning patterns, **the smallest feature** that can be built is around one and a half times the laser beam diameter i.e., around **300 μm** .

in higher resolution systems such as Viper Si2 machine, the laser beam diameter is $70 \pm 15 \mu\text{m}$ and features as thin as $80 \mu\text{m}$ have been built using this spot size.

With smaller spot size capability adds small feature preservation.

This improves accuracy of SLA process

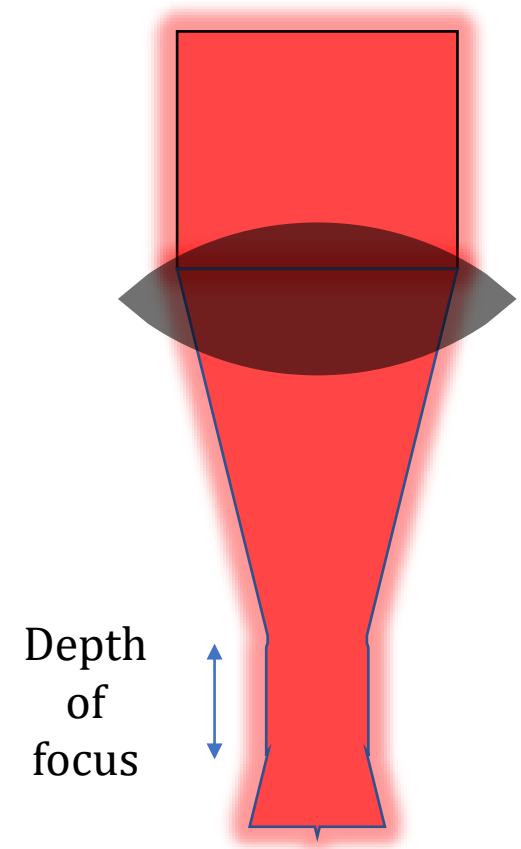


2. Laser beam focus depth:

In the z direction, the **cure depth** of a layer is a **function** of the **focus depth** underneath the resin surface.

The attainable thickness is inversely proportional with the cure depth, hence the depth of focus of the laser beam.

The laser beam is focused beneath the resin surface in order to increase the Rayleigh range and the accurate build platform size. The focus distance is typically 8mm below the resin surface.



3. Layer thickness:

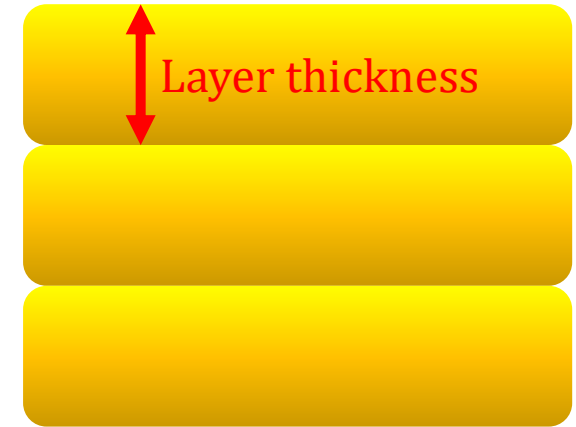
Thickness of one layer is **inversely proportional** to the **depth of focus** of the laser beam under the resin surface.

However, layer thickness is mainly limited by the recoating mechanism and resin viscosity.

The blade (Zephyr) limits the thickness of each layer.

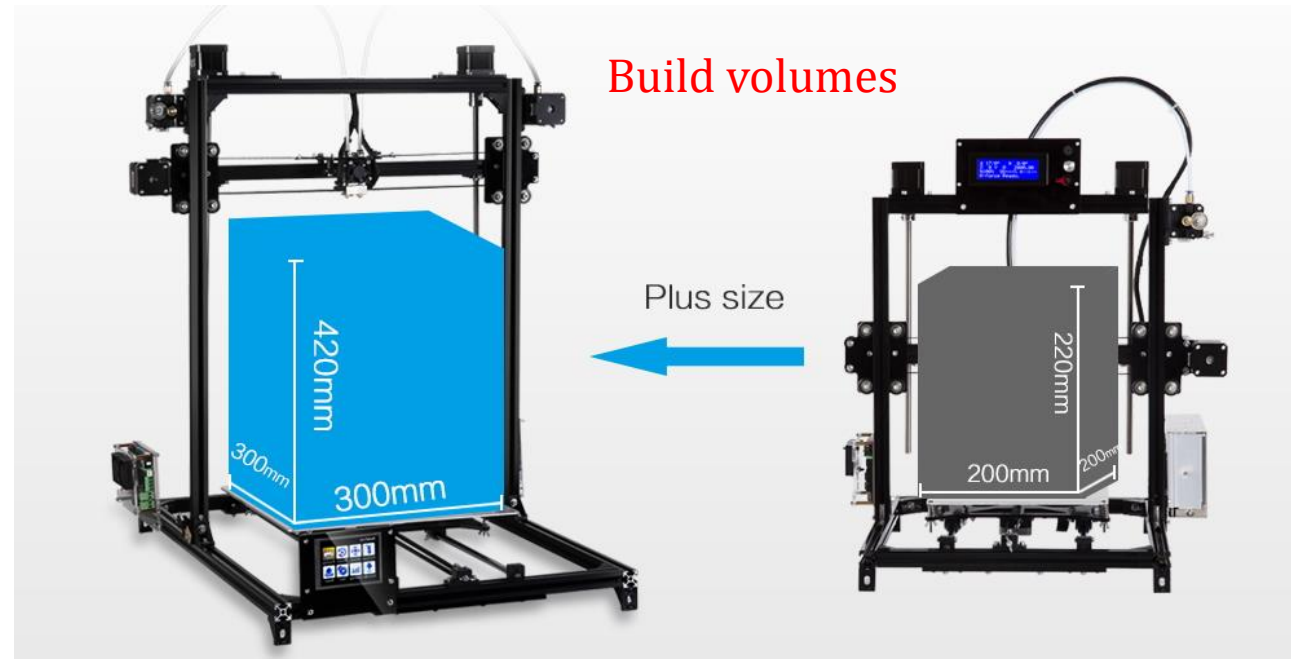
This blade is necessary because SL resins have high viscosities, approximately 180 – 2000 centipoise.

Currently the smallest layer thickness that can be obtained is 25 μm (0.001 inch). The layer thickness affects the surface texture, the accuracy in the z-axis and the build speed.



Advantages of SLA

1. **Unattended building process:** The SLA can be used continuously and unattended round the clock.
2. **Good user support:** The computerized process serves as a good user support.
3. **Wide range of Build volumes:** The different SLA machines have build volumes ranging from small to large to suit the needs of different users.



- 4. **Good accuracy:** The SLA has good accuracy, the built parts are within $\pm 0.1\text{mm}$ accuracy and can thus be used for many application areas.
- 5. **Surface finish:** The SLA can obtain one of the best surface finishes amongst RP technologies.
- 6. The process is of high resolution and capable to build parts with complex details.

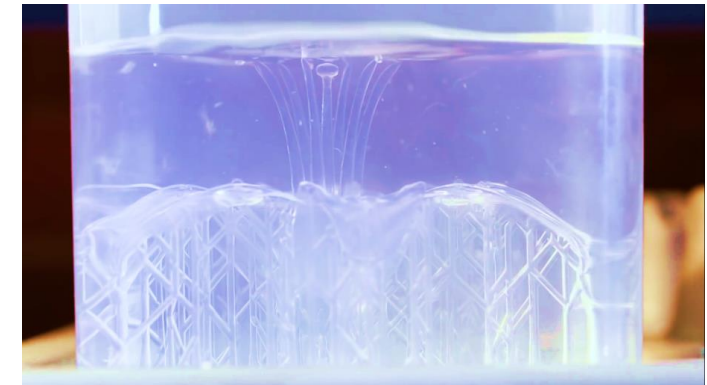


Disadvantages of SLA

1. **Requires support structures:** Structures that have overhangs and undercuts must have supports that are designed and fabricated together with the main structure.
2. **Requires post-processing:** Post-processing includes removal of supports and other unwanted materials, which is tedious, time consuming and can damage the model.
3. **Requires post-curing:** Post-curing may be needed to cure the object completely and ensure the integrity of the structure



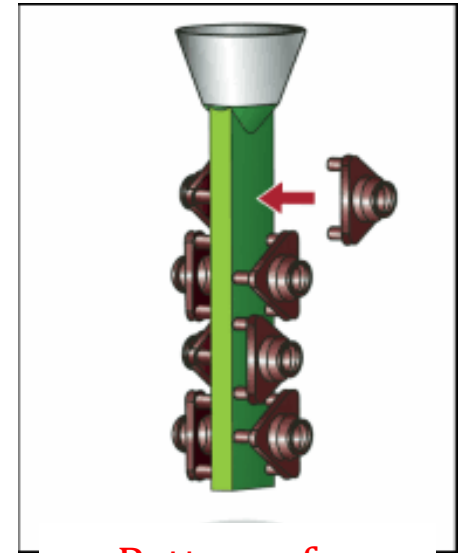
support structures



post-curing

APPLICATIONS

- The SLA technology provides manufacturers with cost justifiable methods for reducing time to market, lowering product development costs, gaining greater control of their design process and improving product design. The range of applications include:
- Models for conceptualization, packaging and presentation.
- Prototypes for design, analysis, verification and functional testing.
- Parts for prototype tooling and low volume production tooling.
- Patterns for investment casting, sand casting and molding.
- Tools for fixture and tooling design, and production tooling



Patterns for
investment
casting